Environment:-

State space:- {Tired, Neutral, Productive}

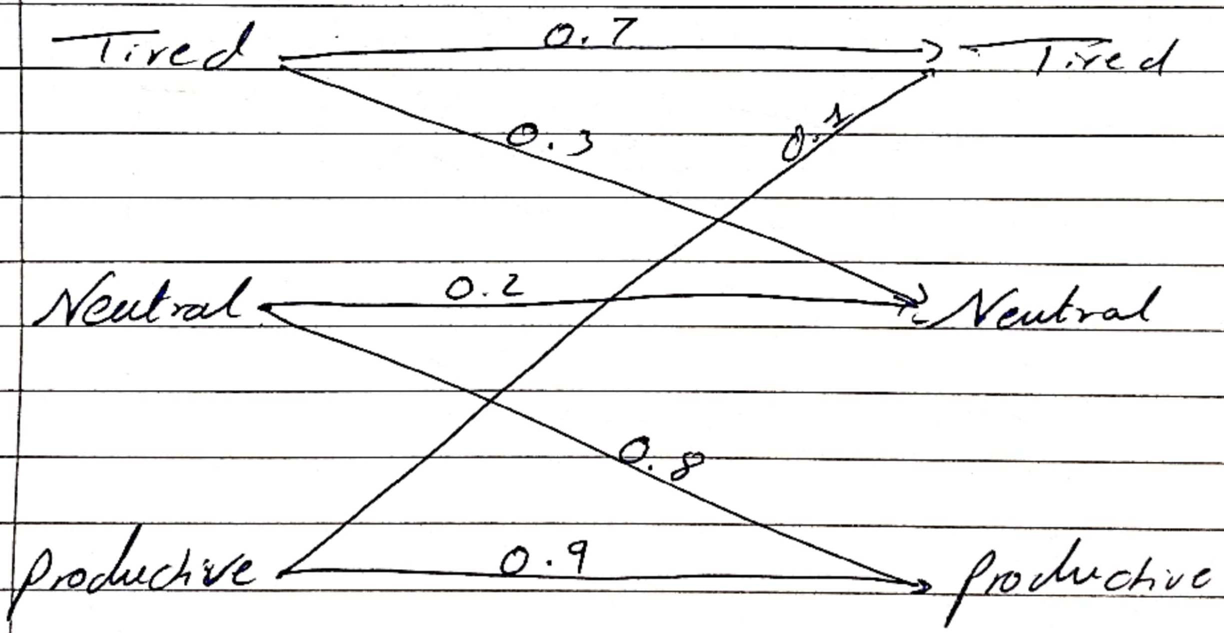
Action space:- {Study, Sleep, ~~Hangout~~}

Transition probability:- Decribes how likely a student is to move to a specific state from current state, After taking a specific action.

$$P(s'|s, a)$$

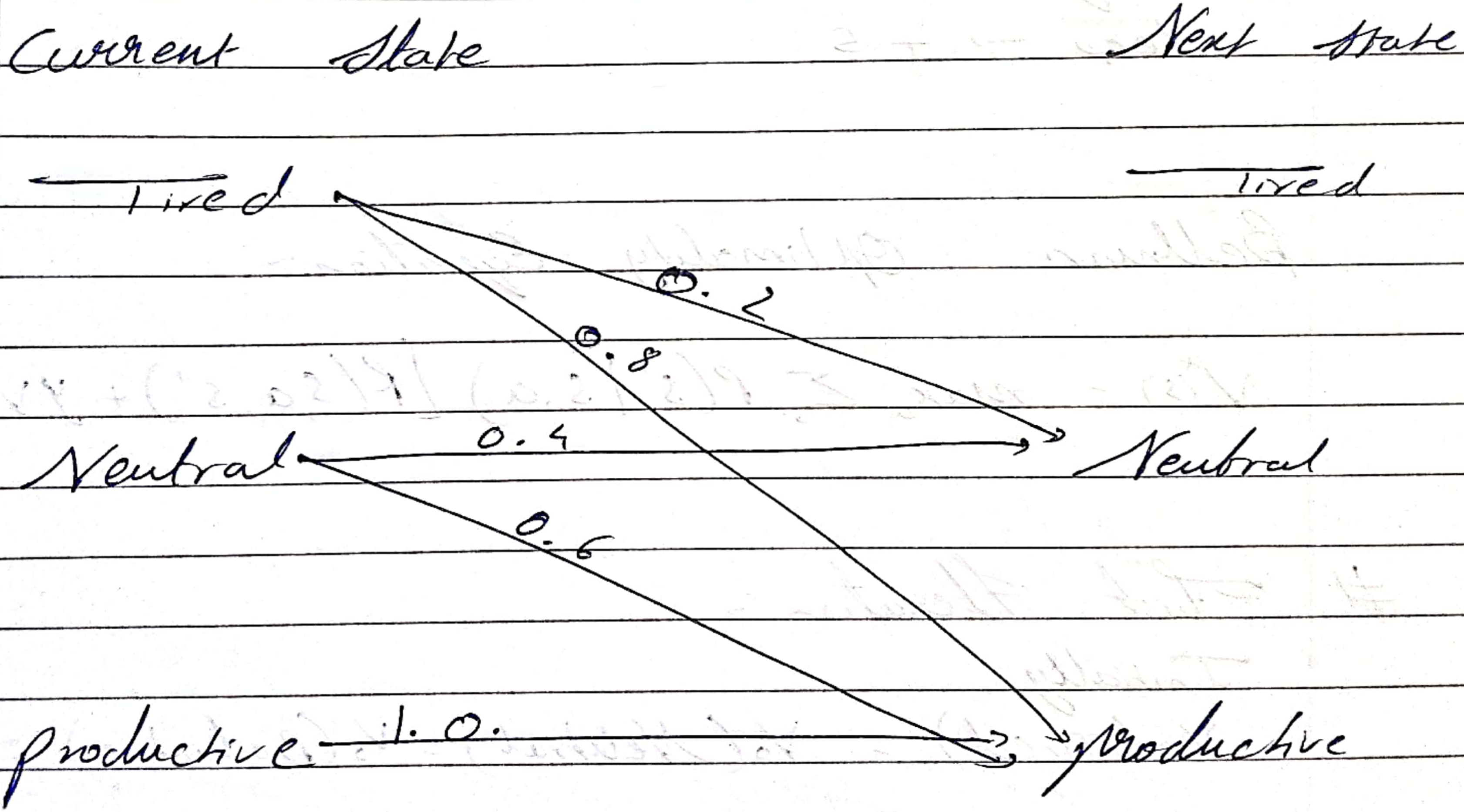
action:- ① Study

Current state   Next state



② Sleep

② Sleep



Current State
Next State
Tired
Tired
0.2
0.8
Neutral
0.4
Neutral
0.6
productive
1.0
productive

__/__/____

Reward :-

Study → +10

Sleep → +5

Bellman Optimality Equation :-

$$V^*(s) = \max_a \sum_{s'} P(s'|s,a)\left[R(s,a,s') + \gamma V^*(s')\right]$$

First Iteration :-

Initially

$V_0(\text{Tired}) = V_0(\text{Neutral}) = V_0(\text{Productive}) = 0$

State : Tired

→ Study

$$\Rightarrow \underbrace{0.7[10 + 0.9\times 0]}_{\substack{\text{Tired} \\ \cancel{\text{productive}}}} + \underbrace{0.3[10 + 0.9\times 0]}_{\substack{\text{Neutral} \\ \cancel{\text{Tired}}}}$$

$= 7 + 3 = 11$

→ Sleep

$$\underbrace{0.8(6+0)}_{\text{productive}} + \underbrace{0.2(6+0)}_{\text{Neutral}}$$

$= 4.8 + 1.2 = 6$

∴ Maximum $V_1(\text{Tired}) = 10$ {action = study}

State :- Neutral

→ Study ⇒ productive + Neutral

$0.8[10+0.9\times 0] + 0.2[10 + 0.9\times 0]$

$= 10$

→ Sleep ⇒ Neutral + productive

$0.6[6] + 0.4[6] = 6$

⇒ Max $V_1(\text{Neutral}) = 10$

State : productive

→ Study => $0.9(10) + 0.1(10) = 10$

→ sleep = $1.0(6) = 6$

$V_1(\text{productive}) = 10$

∴ After Iteration 1

$V_1 = [10, 10, 10]$

II^{nd} Iteration

$V_1(S) = 10$

State : Tired

→ Study :- $0.7[10 + 0.9 \times 10] + 0.3[10 + 0.9 \times 10]$

$\Rightarrow \underbrace{(0.7 + 0.3)}[10 + 0.9 \times 10]$

$\sum \underbrace{P(s'|s,a)} \cdot (R(s,a,s') + \underbrace{\gamma V(s')}_{\cancel{Const} = 9}]$

No dependent term => $[R(s,a,s') + \underbrace{\gamma V(s')}_{Constant = 9}] \sum \underbrace{P(s'|s,a)}_{1}$

$\Rightarrow 10 + 9 = 19$

→ Sleep :- $1[6 + 9] = 15$

$V_2(\text{Tired}) = 19$

State :- ~~productive~~ Neutral

→ study :- $1(10 + 9) = 19$

→ sleep :- $1(6 + 9) = 15$

$\Rightarrow V_2(\text{Neutral}) = 19$

Reward :-

Study → +10

Sleep → +5

Bellman Optimality Equation :-

$$V^*(s) = \max_a \sum_{s'} P(s' \mid s, a)\,[R(s, a, s') + \gamma V^*(s')]$$

First Iteration :-

Initially

$V_0(\text{Tired}) = V_0(\text{Neutral}) = V_0(\text{Productive}) = 0$

State : Tired

→ Study

$$\Rightarrow 0.7\overbrace{[10 + 0.9\times 0]}^{\text{Tired}\ \cancel{\text{productive}}} + 0.3\overbrace{[10 + 0.9\times 10)}^{\text{Neutral}\ \cancel{\text{Tired}}}$$

$$= 7 + 3 = 11$$

→ Sleep

$$0.8\underbrace{(6 + 0)}_{\text{productive}} + 0.2\underbrace{(6 + 0)}_{\text{Neutral}}$$

$$= 4.8 + 1.2 = 6$$

∴ Maximum $V_1(\text{Tired}) = 10$ {action = study}

State :- Neutral

→ Study ⇒ Productive + Neutral

$$0.8[10 + 0.9\times 0] + 0.2[10 + 0.9\times 0]$$

$$= 10$$

→ Sleep ⇒ Neutral + productive

$$0.6[6] + 0.4[6] = 6$$

⇒ Max $V_1(\text{Neutral}) = 10$

__/__/__

State : productive

→ Study ⇒ $0.9(10) + 0.1(10) = 10$

→ Sleep = $1.0(6) = 6$

V_1(productive) = 10

∴ After Iteration 1
$v_1 = [10, 10, 10]$

II^{nd} Iteration
$V_1(S) = 10$

State : Tired

→ Study :- $0.7[10 + 0.9 \times 10] + 0.3[10 + 0.9 \times 10]$

$\Rightarrow (0.7 + 0.3)[10 + 0.9 \times 10]$

$\sum \underbrace{P(s'|s,a)}\cdot[R(s,a,s') + \underbrace{\gamma V(s')}_{\text{Const } 9}]$

No dependent $\Rightarrow [R(s,a,s') + \underbrace{\gamma V(s')}_{\text{Constant} = 9}] \sum \underbrace{P(s'|s,a)}_{1}$

$\Rightarrow 10 + 9 = 19$

→ Sleep :- $1[6 + 9] = 15$

V_2(Tired) = 19

State :- ~~productive~~ Neutral

→ study :- $1(10 + 9) = 19$

→ Sleep :- $1(6 + 9) = 15$

⇒ V_2(Neutral) = 19

__/__/__

state :- productive

$\rightarrow$ Study :- $1[10+9] = 19$

$\rightarrow$ Sleep :- $1[6+9] = 15$

$\therefore$ $V_2(\text{productive}) = 19$

$\therefore$ $V_2(\text{Tired}) = V_2(\text{Neutral}) = V_2(\text{productive}) = 19$

$\Rightarrow$ The above ~~loops~~ is an example of Model-based RL.